QUESTION BANK

DBMS 1-Mark Questions

UNIT – I

1. What is a Database Management System (DBMS)?
2. Define database.
3. What are the advantages of DBMS over file systems?
4. What is data redundancy?
5. Define data inconsistency.
6. What is a data model?
7. Name different types of data models.
8. What is abstraction in DBMS?
9. What are the levels of abstraction in DBMS?
10. Define physical level in DBMS.
11. Define logical level in DBMS.
12. Define view level in DBMS.
13. What is data independence?
14. Define physical data independence.
15. Define logical data independence.
16. What is schema in DBMS?
17. Define instance in DBMS.
18. What are the components of DBMS structure?
19. What is an entity?
20. Define attribute.
21. What is an entity set?
22. What is a relationship?
23. Define relationship set.

24.What is an ER model?

25.What is an ER diagram?

26.What are simple attributes?

27.What are composite attributes?

28.Define multivalued attribute.

29.What is a derived attribute?

30.What is a key attribute?

31.Define weak entity set.

32.What is cardinality ratio?

33.Define participation constraint.

34.What is conceptual database design?

35.What is specialization?

36.Define generalization.

37.What is aggregation in ER model?

38.What is inheritance in ER model?

39.What is an ISA relationship?

40.Define primary key.

UNIT – II

41.What is the relational model?

42.Define relation.

43.What is a tuple?

44.Define attribute in relational model.

45.What is a domain?

46.Define degree of relation.

47.Define cardinality of relation.

48.What is an integrity constraint?

49.Define entity integrity constraint.

50.Define referential integrity constraint.

51.What is a candidate key?

52.Define super key.

53.What is an alternate key?

54.Define foreign key.

55.What is relational algebra?

56.Name the basic operations in relational algebra.

57.What is selection operation?

58.Define projection operation.

59.What is union operation in relational algebra?

60.Define set difference operation.

61.What is Cartesian product?

62.Define join operation.

63.What is theta join?

64.Define equi-join.

65.What is natural join?

66.Define division operation.

67.What is tuple relational calculus?

68.Define domain relational calculus.

69.What is a query language?

70.Define view in DBMS.

71.What are advantages of views?

72.What is logical database design?

73.Define relational schema.

74.What is meant by altering a table?

75.What is meant by dropping a table?

UNIT – III

76. What is SQL?
77. Expand DDL.
78. Expand DML.
79. Expand DCL.
80. Expand TCL.
81. What is the basic form of SQL query?
82. What is the use of SELECT statement?
83. Define WHERE clause.
84. What is ORDER BY clause?
85. Define GROUP BY clause.
86. What is HAVING clause?
87. What is UNION operator?
88. Define INTERSECT operator.
89. What is EXCEPT operator?
90. Define nested query.
91. What are aggregate functions in SQL?
92. Name any four aggregate functions.
93. What is NULL value?
94. Define integrity constraints in SQL.
95. What is NOT NULL constraint?
96. Define UNIQUE constraint.
97. What is CHECK constraint?
98. Define DEFAULT constraint.
99. What is a trigger?
100. Define active database.

101. What is redundancy in DBMS?

102. What problems are caused by redundancy?

103. Define decomposition in DBMS.

104. What is lossless decomposition?

105. What is lossy decomposition?

106. Define functional dependency.

107. What is trivial functional dependency?

108. What is non-trivial functional dependency?

109. Define full functional dependency.

110. What is partial dependency?

111. Define transitive dependency.

112. What is normalization?

113. What is First Normal Form (1NF)?

114. What is Second Normal Form (2NF)?

115. What is Third Normal Form (3NF)?

116. What is BCNF?

117. What is the difference between 3NF and BCNF?

118. Define multivalued dependency.

119. What is Fourth Normal Form (4NF)?

120. What is Fifth Normal Form (5NF)?

UNIT – IV (Transactions & Concurrency Control)

121. What is a transaction?
122. Define transaction state.
123. Name transaction states.
124. What is atomicity?
125. What is consistency in DBMS?
126. What is isolation?
127. What is durability?
128. What is ACID property?
129. What is concurrent execution?
130. What is serial schedule?
131. Define serializability.
132. What is conflict serializability?
133. What is view serializability?
134. Define recoverable schedule.
135. What is cascading rollback?
136. What is strict schedule?
137. What is lock-based protocol?
138. Define shared lock.
139. Define exclusive lock.
140. What is two-phase locking (2PL)?
141. What is timestamp ordering?
142. What is validation-based protocol?
143. What is deadlock?
144. What is multiple granularity?
145. What is intention lock?

146. What is recovery in DBMS?

147. What is log-based recovery?

148. What is undo operation?

149. What is redo operation?

150. What is checkpoint in DBMS?

UNIT – V (File Organization & Indexing)

151. What is file organization?

152. What is data storage in DBMS?

153. What is a file?

154. What is record?

155. What is field in DBMS?

156. Define file organization methods.

157. What is heap file organization?

158. What is sequential file organization?

159. What is indexed file organization?

160. What is hashed file organization?

161. What is indexing?

162. Why indexing is used in DBMS?

163. What is primary index?

164. What is secondary index?

165. What is cluster index?

166. Define dense index.

167. Define sparse index.

168. What is index key?

169. What is multilevel indexing?

170. What is B-tree?

171. What is B+ tree?

172. What are the properties of B-tree?

173. What is difference between B-tree and B+ tree?

174. What is hash based indexing?

175. What is hashing?

176. What is hash function?

177. What is collision in hashing?

178. What is open hashing?

179. What is closed hashing?

180. What is linear probing?

181. What is quadratic probing?

182. What is bucket in hashing?

183. What is overflow in file organization?

184. What is primary file organization?

185. What is secondary storage?

186. What is disk storage?

187. What is access time?

188. What is seek time?

189. What is rotational delay?

190. What is transfer time?

191. What is clustered file organization?

192. What is comparison of file organizations?

193. Which file organization is fastest?

194. Which file organization uses least memory?

195. What is indexing advantage?

196. What is indexing disadvantage?

197. What is database performance tuning?

198. What is sequential access?

199. What is random access?

200. What is direct access method?

201. What is database security?

202. What is authorization in DBMS?

203. What is authentication?

204. What is database recovery?

205. What is backup in DBMS?

206. What is data warehousing?

207. What is data mining?

208. What is OLTP?

209. What is OLAP?

210. What is distributed database?

211. What is client-server architecture?

212. What is three-tier architecture?

213. What is query optimization?

214. What is database tuning?

215. What is metadata?

216. What is catalog in DBMS?

217. What is DBA?

218. What are responsibilities of DBA?

219. What is data dictionary?

220. What is database engine?

221. What is schema independence?

222. What is physical storage?

223. What is buffer management?

224. What is cache in DBMS?

225. What is transaction log?

226. What is concurrency control?

227. What is integrity constraint violation?

228. What is normalization anomaly?

229. What is deadlock prevention?

230. What is database lifecycle?

DBMS – 5 MARK IMPORTANT QUESTIONS

UNIT – I: Database Basics & ER Model

1. Explain database system applications and historical perspective.
2. Differentiate between file system and DBMS.
3. Explain data model and its types.
4. Explain levels of abstraction in DBMS with diagram.
5. Explain data independence in detail.
6. Explain structure of DBMS with diagram.
7. Explain database design and ER diagrams.
8. Explain entities, attributes, and entity sets with examples.
9. Explain types of attributes in ER model.
10. Explain relationships and relationship sets.
11. Explain additional features of ER model (specialization, generalization, aggregation).
12. Explain conceptual design using ER model with example.

UNIT – II: Relational Model & Relational Algebra

1. Explain relational model and its structure.
2. Explain integrity constraints over relations.
3. Explain entity integrity and referential integrity constraints.
4. Explain enforcing integrity constraints in DBMS.
5. Explain querying relational data with examples.
6. Explain logical database design.
7. Explain views in DBMS with advantages.
8. Explain SQL operations for altering and destroying tables/views.
9. Explain relational algebra operations (selection, projection, union, difference).
10. Explain different types of join operations.
11. Explain tuple relational calculus.
12. Explain domain relational calculus.
13. Compare relational algebra and relational calculus.

UNIT – III: SQL & Normalization

1. Explain basic structure of SQL query.
2. Explain UNION, INTERSECT, EXCEPT with examples.
3. Explain nested queries in SQL.
4. Explain aggregate functions in SQL.

5. Explain NULL values and their handling.
6. Explain triggers and active databases.
7. Explain complex integrity constraints in SQL.
8. Explain redundancy and its problems.
9. Explain decomposition in DBMS.
10. Explain functional dependency with examples.
11. Explain 1NF with example.
12. Explain 2NF with example.
13. Explain 3NF with example.
14. Explain BCNF with example.
15. Explain lossless join decomposition.
16. Explain multivalued dependency.
17. Explain 4NF with example.
18. Explain 5NF with example.

UNIT – IV: Transactions & Concurrency Control

1. Explain transaction concept and states.
2. Explain ACID properties with examples.
3. Explain implementation of atomicity and durability.
4. Explain concurrent executions in DBMS.
5. Explain serializability with example.
6. Explain recoverability of schedules.

7. Explain isolation and its implementation.
8. Explain testing for serializability.
9. Explain lock-based protocols.
10. Explain timestamp-based protocols.
11. Explain validation-based protocols.
12. Explain multiple granularity locking.
13. Explain deadlock and its handling methods.
14. Explain log-based recovery.
15. Explain recovery with concurrent transactions.
16. Explain undo and redo operations.
17. Explain checkpointing technique.
18. Explain different transaction schedules.

UNIT – V: File Organization & Indexing

1. Explain data storage on external storage.
2. Explain file organization and its types.
3. Explain heap file organization.
4. Explain sequential file organization.
5. Explain indexed file organization.
6. Explain hashed file organization.
7. Compare different file organizations.

8. Explain primary, secondary, and cluster indexes.
9. Explain dense and sparse indexing.
10. Explain index data structures.
11. Explain B-tree with example.
12. Explain B+ tree with example.
13. Compare B-tree and B+ tree.
14. Explain hash-based indexing.
15. Explain hash functions and collisions.
16. Explain linear probing method.
17. Explain quadratic probing method.
18. Explain performance of indexing techniques.